

ARCHITECTURE OF THE GSM

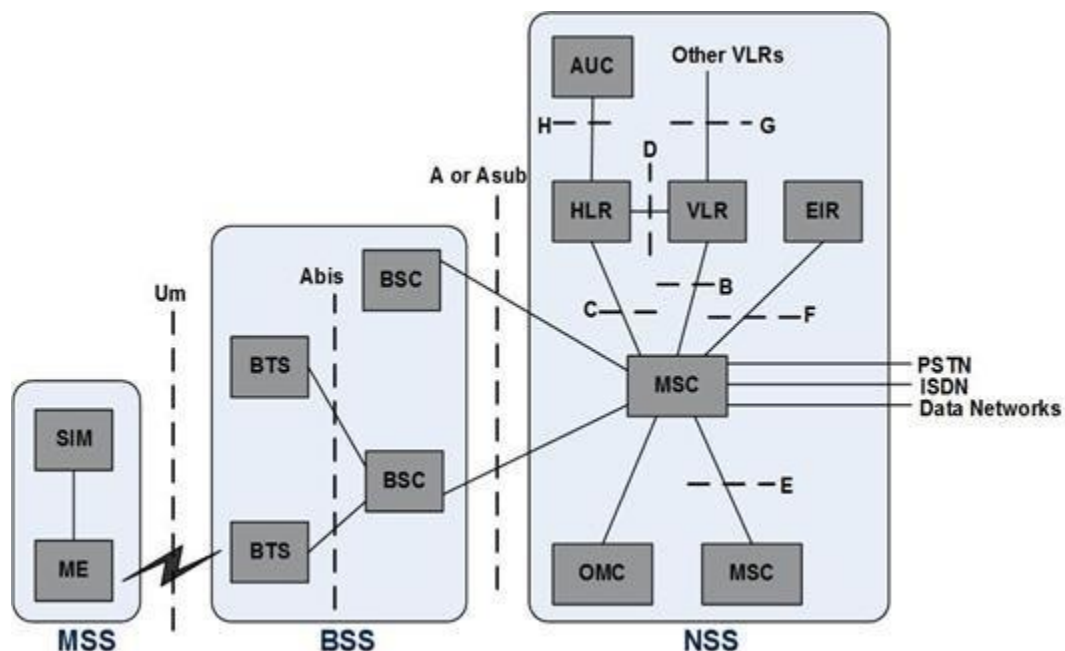
- GSM is a public land mobile network(PLMN)
- Several providers set-up mobile networks following the GSM standard within each country

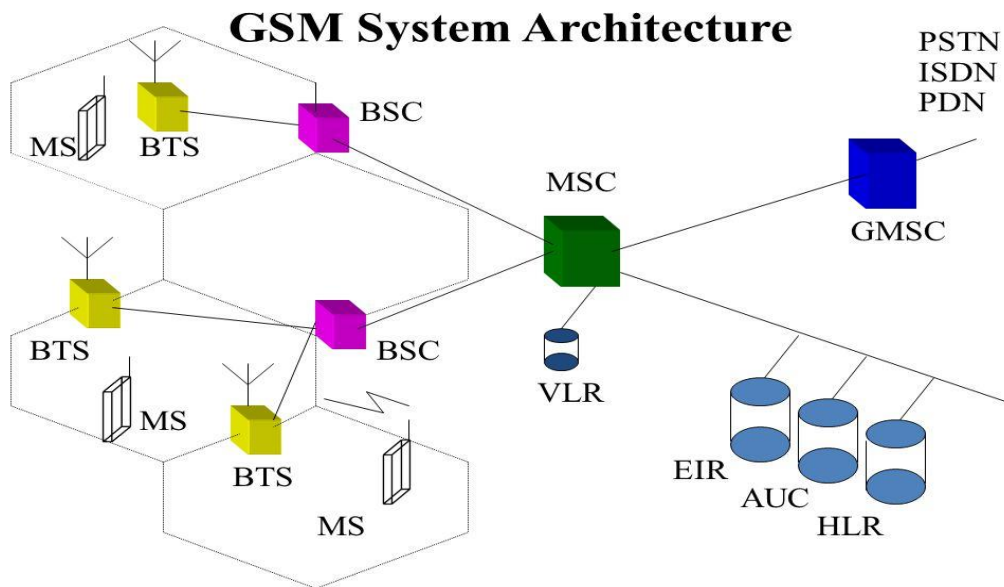
Components

- MS (Mobile Station)
- BS(Base Station)
- MSC(Mobile Switching Centre)
- LR (Location Register)

Sub-Systems

- RSS (radio subsystem): Covers all radio aspects
- NSS(Network and Switching subsystem): Call forwarding, handover and switching
- OSS(Operation subsystem): management of network





A. Radio subsystem

As the name implies, the radio subsystem (RSS) comprises all radio specific entities, i.e., the mobile stations (MS) and the base station subsystem (BSS). Figure 4.4 shows the connection between the RSS and the NSS via the A interface (solid lines) and the connection to the OSS via the O interface. The A interface is typically based on circuit-switched PCM-30 systems (2.048 Mbit/s), carrying up to 30 64 kbit/s connections, whereas the O interface uses the Signalling System No. 7 (SS7) based on X.25 carrying management data to/from the RSS.

B. Base station subsystem (BSS): A GSM network comprises many BSSs, each controlled by a base station controller (BSC). The BSS performs all functions necessary to maintain radio connections to an MS, coding/decoding of voice, and rate adaptation to/from the wireless network part. Besides a BSC, the BSS contains several BTSs.

a) Base transceiver station (BTS):

A BTS comprises all radio equipment, i.e., antennas, signal processing, amplifiers necessary for radio transmission. A BTS can form a radio cell or, using sectorized antennas, several cells and is connected to MS via the **Um interface** (ISDN U interface for mobile use), and to the BSC via the **Abis interface**. The Um interface contains all the mechanisms necessary for wireless transmission (TDMA, FDMA etc.). The Abis interface consists of 16 or 64 kbit/s connections. A GSM cell can measure between some 100 m and 35 km depending on the environment (buildings, open space, mountains etc.) but also expected traffic.

b) Base station controller (BSC):

The BSC basically manages the BTSs. It reserves radio frequencies, handles the handover from one BTS to another within the BSS, and performs paging of the MS. The BSC also multiplexes the radio channels onto the fixed network connections at the A interface.

C. Mobile station (MS):

The MS comprises all user equipment and software needed for communication with a GSM network.

Subscriber identity module (SIM)- stores all user-specific data that is relevant to GSM(charging and authentication) The SIM card contains many identifiers and tables, such as card-type, serial number, a list of subscribed services, a personal identity number (PIN), a PIN unblocking key (PUK), an authentication key K_i , and the international mobile subscriber identity (IMSI). The MS stores dynamic information while logged onto the GSM system, such the cipher key K_c and the location information consisting of a temporary mobile subscriber identity (TMSI) and the location area identification (LAI).

Device- Identified via the international mobile equipment identity (IMEI), (Device specific mechanisms such as theft protection uses IMEI)

Typical MSs for GSM 900 have a transmit power of up to **2 W**, whereas for GSM 1800 **1 W** is enough due to the smaller cell size.

D. Network and switching subsystem

The “heart” of the GSM system is formed by the network and switching subsystem (NSS). The NSS connects the wireless network with standard public networks, performs handovers between different BSSs, comprises functions for worldwide localization of users and supports charging, accounting, and roaming of users between different providers in different countries. The NSS consists of the following switches and databases:

a) Mobile services switching center (MSC):

The VLR associated to each MSC is a dynamic database which stores all important information needed for the MS users currently in the LA that is associated to the MSC (e.g., IMSI, MSISDN, HLR address). If a new MS comes into an LA the VLR is responsible for, it copies all relevant information for this user from the HLR. This hierarchy of VLR and HLR avoids frequent HLR updates and long-distance signaling of user information.

b) Home location register (HLR):

The HLR is the most important database in a GSM system as it stores all user-relevant information.

- This comprises **static information**, such as :

- the mobile subscriber ISDN number (MSISDN), subscribed services (e.g., call forwarding, roaming restrictions, GPRS) (Mobile Station International Subscriber Directory Number - (MSISDN)
- The international mobile subscriber identity (IMSI).
- Dynamic information is also needed, e.g.,
 - the current location area (LA) of the MS,
 - the mobile subscriber roaming number (MSRN),
 - the current VLR and MSC.

As soon as an MS leaves its current LA, the information in the HLR is updated. This information is necessary to localize a user in the worldwide GSM network. All these user-specific information elements only exist once for each user in a single

- HLR, supports charging and accounting.
- HLRs manages data for several million customers and contain highly specialized data bases which must fulfill certain real-time requirements to answer requests within certain time-bounds.

c) Visitor location register (VLR):

The VLR associated to each MSC is a dynamic database which stores all important information needed for the MS users currently in the LA that is associated to the MSC (e.g., IMSI, MSISDN, HLR address). If a new MS comes into an LA the VLR is responsible for, it copies all relevant information for this user from the HLR. This hierarchy of VLR and HLR avoids frequent HLR updates and long-distance signaling of user information. The typical use of HLR and VLR for user. Some VLRs in existence, are capable of managing up to one million customers.

D. Operation subsystem

The third part of a GSM system, the operation subsystem (OSS), contains the necessary functions for network operation and maintenance. The OSS possesses network entities of its own and accesses other entities via SS7 signaling. The following entities have been defined:

a) Operation and maintenance center (OMC):

The OMC monitors and controls all other network entities via the O interface (SS7 with X.25). Typical OMC management functions are traffic monitoring, status reports of network entities, subscriber and security management, or accounting and billing. OMCs use the concept of telecommunication management network (TMN) as standardized by the ITU-T.

b) Authentication Centre (AuC):

As the radio interface and mobile stations are particularly vulnerable, a separate AuC has been defined to protect user identity and data transmission. The AuC contains the algorithms for

authentication as well as the keys for encryption and generates the values needed for user authentication in the HLR. The AuC may, in fact, be situated in a special protected part of the HLR.

c) Equipment identity register (EIR):

The EIR is a database for all IMEIs, i.e. it stores all device identifications registered for this network. As MSs are mobile, they can be easily stolen. With a valid SIM, anyone could use the stolen MS. The EIR has a blacklist of stolen (or locked) devices. In theory an MS is useless as soon as the owner has reported a theft. Unfortunately, the blacklists of different providers are not usually synchronized and the illegal use of a device in another operator's network is possible. The EIR also contains a list of valid IMEIs (white list), and a list of malfunctioning devices (gray list).